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Ultrasound real-time imaging of crack propagation during ultrasonic fatigue tests

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ABSTRACT

This Letter describes a real-time surface inspection method based on ultrasonic synthetic imaging using Rayleigh waves. The proposed method can be employed to monitor crack propagation during ultrasonic fatigue tests. The setup, which is highly sensitive to cracks, enables the imaging of a large surface (tens of mm^2), with spatial resolutions of several tens of micrometers and an imaging frequency of up to 50 frames per second. The micrometric resolution allows cracks to be investigated at the grain level, while the sample imaging enables high-velocity crack propagation to be accurately resolved. These two domains have rarely been investigated to date because of experimental limitations. This work paves the way for a new type of laboratory-scale fatigue monitoring based on crack propagation, leading to better comprehension of the associated physical phenomena.

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Ultrasonic fatigue testing machines have undergone significant developments in recent decades. At present, they are powerful experimental tools for characterizing the fatigue behavior of materials over a very large number of cycles, with testing times reduced by a factor of up to 1000 compared with conventional hydraulic machines.[1,2] For example, at a testing frequency of 20 kHz, one test over 10^9 cycles takes half a day, and fatigue curves relating the stress amplitude to the number of cycles can be developed over up to a billion cycles in just a few days. Ultrasonic fatigue machines also enable fatigue crack propagation to be studied, as suggested by Tien in 1975.[3] In particular, they can investigate near-threshold propagation laws and very low propagation speeds, which can drop to 1×10^{-13} m/cycle or less. This technique has been used by various researchers to study frequency,[4,5] environmental,[4,6–8] average,[4,9] and crystallographic texture[5] effects on near-threshold fatigue crack propagation. One of the main challenges in such studies is to estimate the stress intensity factor amplitude. Initially determined using equations derived from static loading, several studies have incorporated the vibration aspect of the loading, thus considering the effects of inertia.[10–12] The amplitude of the stress intensity factor is then estimated from the displacement field near the crack tip. The question of how the mesh resolution affects the estimation of the stress intensity factor remains unresolved.

In general, fatigue crack propagation monitoring is performed by observing the surface using an optical microscope. A moderate numerical aperture allows spatial resolutions of less than 1 μm to be achieved, but the technique requires the crack tip to be precisely identified, which generally requires careful polishing of the surface. Two images taken before and after a block of cycles are used to determine the initial and final crack lengths, from which the propagation increment and growth rate can be determined. This is a laborious process that severely limits the number of measurement points during the test. Continuous real-time measurements using video recording could overcome this issue, but these have seldom been reported in the literature,[13] mainly because of the highly restricted field of view at the required resolutions. Thus, dynamical monitoring of crack propagation remains a considerable challenge, although it could allow high propagation speeds (beyond 1×10^{-7} m/cycle) to be investigated, corresponding to high-amplitude stress intensity factors.

Thermography provides another method for detecting and monitoring fatigue cracks.[14] However, this approach faces the same dilemma as visible optical approaches, requiring a trade-off between the size of the field of view and the resolution. For instance, the *in situ* detection of fatigue cracks through thermal radiation monitoring has been proposed, but this method requires the cracks to be sufficiently long (of the order of mm).[15] A low-cost approach has also been proposed, but this requires an important averaging of images, strongly reducing the effective sampling frequency.[16]

Ultrasonic fatigue testing machines are powerful instruments for investigating the fatigue crack propagation behavior of materials at high stress intensity factors. Unfortunately, they require dedicated *in*

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situ monitoring solutions combining high spatial resolution and adapted temporal resolutions. The first criterion is crucial for studying very slow propagation speeds and near-threshold crack growth, while the latter is necessary to track high propagation speeds and investigate material behaviors at high-amplitude stress intensity factors.

X-ray phase-contrast imaging using synchrotron radiation can be employed to monitor crack propagation with a very high sampling frequency (of the order of MHz).[17] However, this approach is beyond the scope of lab-scale equipment and is consequently difficult to access. On the contrary, elastic waves are widely used for nondestructive testing (NDT), particularly to detect inhomogeneities with a strong impedance mismatch, such as cracks or voids. The advantages of elastic waves include their long propagation distances and absence of physical disturbance due to their low energy. This is why they are excellent candidates for probing crack propagation non-invasively during fatigue testing. A fatigue crack propagation monitoring technique has been used to monitor the transmission and reflection coefficients of the S_0 Lamb mode between pairs of piezoelectric transducers placed on either side of a crack.[18] However, this technique requires the fatigue testing to be paused during the ultrasonic propagation and has only been demonstrated for a loading frequency of several Hz. Bulk elastic waves with a central frequency of 5 MHz have been used to monitor crack propagation,[19] with a 22-element transducer array employed to perform the total focusing method (TFM) based on full matrix capture (FMC) datasets. Although similar to the technique described in this article, this approach requires a delicate transducer installation and was only used between loading cycle blocks.

In 2020, Ducousso and Reverdy proposed a real-time NDT ultrasonic imaging method for metallic surfaces, which can precisely characterize complex surface cracks at an imaging rate of 27 frames per second (fps).[20] This method uses a phased-array (PA) probe fixed on a wedge and placed at the critical angle to generate surface waves. Synthetic imaging allows well-defined crack images to be realized. This technique has been used to image isolated corrosion pitting defects with a diameter as small as 100 μm distributed on a surface larger than 45 × 12 mm.[21] Similar results could be achieved with a relatively high signal-to-noise ratio (several tens of dB). This paper describes the application of this technique to monitor crack propagation at 50 fps during fatigue testing. Notably, we were able to record the whole propagation in one go with both high spatial and temporal resolutions with respect to the propagation dynamics. This paper presents an original analysis, based on a time integration of Paris' law, to fit the experimental data and extract material parameters. The experimental setup and aspects are first described, and then the signal processing approach is explained. Finally, the results are discussed.

Fatigue stress was applied by an ultrasonic fatigue machine classically used for determining stress amplitude vs number of cycles curves.[1] The machine consists of a Branson CR20 converter with a vibration frequency of 20 kHz, a 1:1.5 amplification booster fixed on the converter, and a horn with an amplification coefficient of 3. The R-ratio of the cycling is equal to −1. The system was calibrated by measuring the displacement amplitude at the end of the horn with a laser vibrometer (Polytech) as a function of the voltage amplitude applied to the converter. The fatigue machine can produce a vibration amplitude of 5–50 μm at the end of the horn. During the experiments, the load amplitude was gradually increased to reach the crack propagation threshold. The vibration amplitude was then held constant throughout the whole propagation. As soon as the crack became too long, causing an excessive drop in frequency, the machine stopped automatically. The material under investigation was a 2219 aluminum alloy that had undergone T851 heat treatment. The specimens were parallelepipeds with a notch at half-length that were screwed to the end of the horn. The thickness and width of the specimens were fixed at 5 and 20 mm, respectively. The length L was determined to obtain a frequency f_0 for the first longitudinal vibration eigenmode that was close to the vibration frequency of the ultrasonic fatigue machine. Under a one-dimensional harmonic model, the specimen length can be expressed as $L = c/(2f_0)$, where $c = \sqrt{E/\rho}$ is the velocity, $E = 73.1$ GPa is Young's modulus, and $\rho = 2730\,\mathrm{kg\,m^{-3}}$ is the density. Thus, a frequency of 20 kHz gives a length of 128.2 mm. The geometry of a typical specimen is shown in Fig. 1.

Crack propagation monitoring was performed using the method proposed by Ducousso *et al.*[20,21] and illustrated in Fig. 1. This technique is based on Rayleigh waves, generated and detected by a single PA system made of Q piezoelectric elements, and uses either TFM imaging or plane wave imaging (PWI). The latter was chosen for this study to enable high-frequency imaging. PWI is based on the successive generation of N_θ plane waves for different incidence angles. For each generation, labeled $n \in [1, N_\theta]$, backscattered signals $s(t)$ are measured by each of the PA elements, labeled $j \in [1, Q]$. The region of interest (ROI) is meshed (see Fig. 1), and for each point P, the time-of-flight t^P between this point and each of the PA elements is calculated from the ultrasonic velocity of the medium. A synthetic image is obtained by a coherent summation of the $N_\theta \times Q$ signals for every point P using the expression[22]

$$A(P) = \sum_{n=1}^{N_\theta} \sum_{q=1}^{Q} s_{qj}\left(t_n^P + t_q^P\right). \tag{1}$$

The PWI method is similar to FMC/TFM, which is currently the reference imaging technique in NDT, but allows high-frequency ultrasound imaging with high image quality from fewer signals ($N_\theta \times Q$ instead of $Q \times Q$).

While the PA usually emits longitudinal waves, Rayleigh waves were obtained using mode conversion at the interface between the wedge and the sample. The wedge angle was calculated by considering

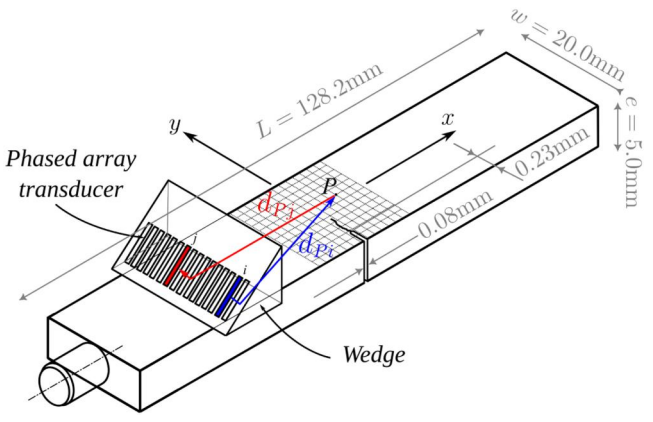


FIG. 1. Geometry of the specimen and imaging mesh.

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the wave velocities and Snell's law.[20] In this study, the wedge was made of rexolite, in which the longitudinal velocity is $v_L^w = 2350\,\mathrm{m\,s^{-1}}$. The velocity of the Rayleigh waves in the specimen was experimentally measured to be $v_R^s = 3215\,\mathrm{m\,s^{-1}}$, leading to an angle of 47°. Rayleigh waves generated using the wedge method are strongly directive and propagate in front of the wedge (along x in Fig. 1). Thus, the PA (Imasonic, France) probe fixed to the wedge was placed a few centimeters above the crack propagation zone, as shown in Fig. 1. The PA consisted of 128 elements, with an elementary pitch of 0.18 mm and a central frequency of $f = 15$ MHz. The dimension along the passive plane was 5 mm. To avoid fatigue perturbation, the PA probe was not placed directly on the sample. Instead, a mechanical system suspended the PA above the specimen, and acoustic transmission was ensured using a coupling liquid. The absence of perturbation was checked by monitoring the frequency resonance of the sample during the installation setup process.

The PA driver (Panther, Eddify Technology) was optimized to perform PWI in real time. We used $N_\theta = 15$ angles, ranging from −20° to +20°. The ROI had a width of $D_x = 10$ mm and a length of $D_y = 20$ mm. The position of the ROI was adjusted in such a way that the initial notch was located at $D_x/2$ and $y = 2$ mm. The pixel size was fixed at $\Delta = 0.025$ mm, chosen to be smaller than the known achievable resolution (evaluated in a previous study to be close to 100 μm for the same equipment.[21]) Considering the working frequency and the Rayleigh velocity, the wavelength was close to 215 μm. With those parameters, the maximal achievable image acquisition frequency was 50 fps. This value could be improved by changing the acquisition scheme or by reducing the size of the ROI, but was judged to be appropriate for the initial demonstration because the maximal expected crack propagation velocity was only a few tens of μm/ms.

Over the course of an entire fatigue test, a total of N_T images were acquired. There are several ways of analyzing such data, but a simple approach was considered in this study to correlate the physical phenomena with the data analysis. For each image, a Hilbert transform was applied along the x-axis to recover the envelope of the signal. An example is displayed in Fig. 2(a), in which the crack is located at around $x = 5$ mm. The signal between $y = 1.5$ mm and $y = 2.5$ mm is the notch, visible throughout the experiment. For $y > 2.5$ mm, the crack was initiated by the fatigue experiment. As the Rayleigh waves are totally reflected on the free surface, its width in the ultrasonic image is not related to the physical crack opening, but to the duration of the elastic pulse. The notch position close to the specimen boundary meant that fewer elements contributed to its effective imaging in comparison to the fatigue crack, explaining the lower amplitude. The notch is, however, invariant over the course of the whole experiment, and can thus be used as a reference.

As illustrated in Fig. 2(a), the crack mainly propagated along the y direction due to the one-dimensional nature of the fatigue testing stress field. Therefore, ultrasonic images cropped to a smaller ROI, as indicated in Fig. 2(a) by white dotted lines. Taking advantage of this unidirectionality, summation can be performed along the x-axis in the reduced ROI. The resulting one-dimensional signal, displayed in Fig. 2(b), expresses the intensity of the reflected acoustic signal as a function of y. As long as the crack propagation is collinear with the latter, no information is lost during this operation. For longer cracks, however, the symmetry of the sample is broken, and additional vibration modes can be excited. These can lead to off-axis propagation.

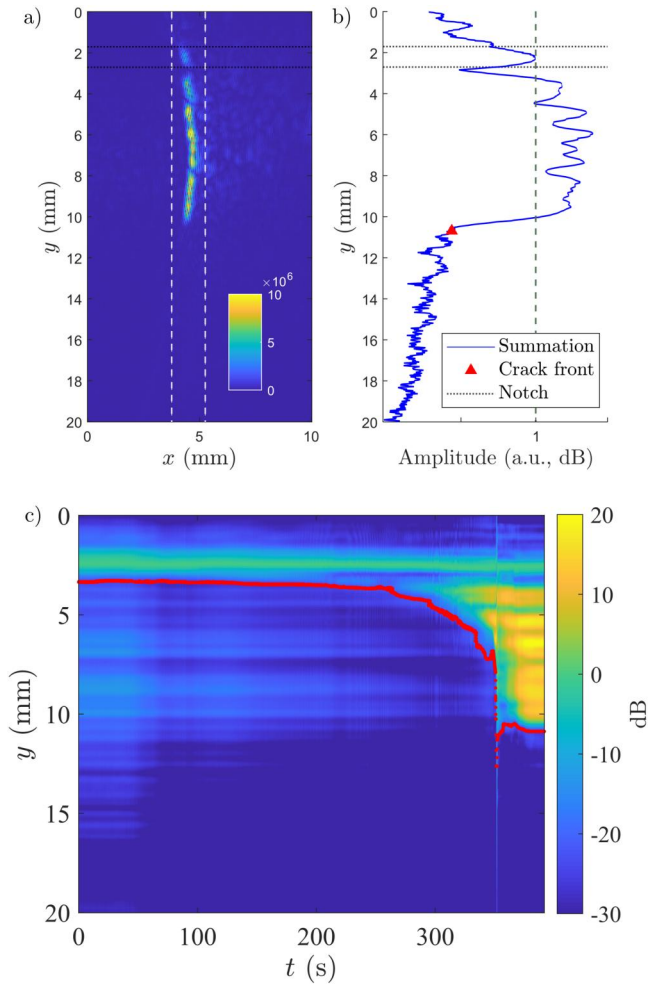


FIG. 2. (a) TFM image at $t = 378.2$ s. (b) Summation of (a) along the x-axis. (c) Time variation of the summations.

The proposed approach then becomes inapplicable; data corresponding to such propagation were not used in the current study.

Performing the operation for all N_T images allowed us to generate a time-resolved image of the crack propagation, as illustrated in Fig. 2(c). That image has had amplitude normalization with respect to the notch signal applied. Moreover, the image is presented in dB scale. A signal-to-noise ratio of a few tens of dB has been recovered.[21] For each time-profile, the crack tip position was determined by applying a threshold at −18 dB [triangle in Fig. 2(b)]. This allowed the crack signal to be separated from the noise. Smoothing over three successive time steps was applied to reduce the influence of noise. The crack-tip position as a function of time is indicated by the red markers in Fig. 2(c). At first, no clear propagation occurred, but after 50 s, which corresponds to 10^6 cycles, the crack began to propagate. The initially linear propagation displayed exponential behavior after around 250 s, i.e., 5×10^6 cycles. The total duration of the test was just over 4 min. To model the results, Rice's J-integral was numerically calculated along a circular path centered at the crack tip,[23] allowing us to

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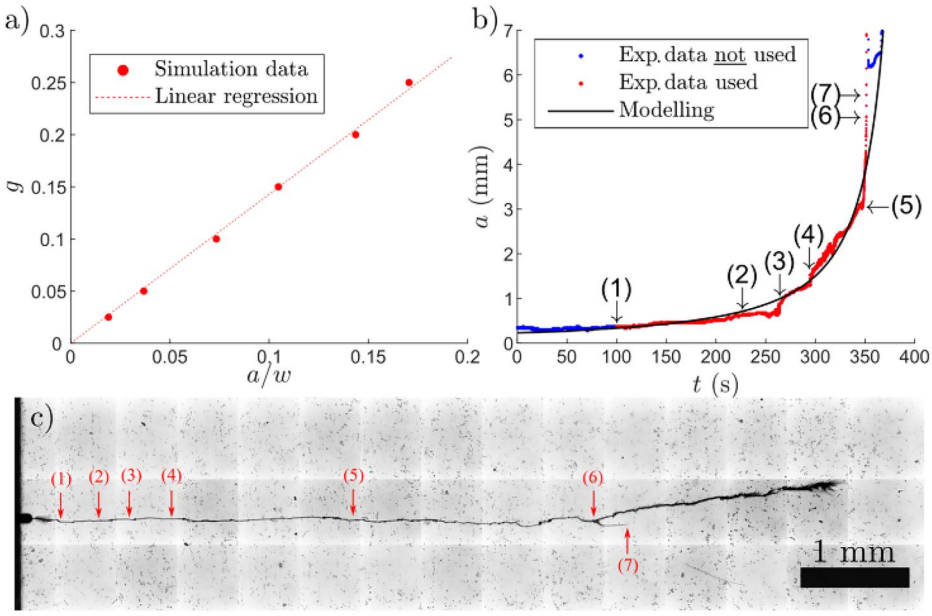


FIG. 3. (a) Identification of the g function. (b) Experimental monitoring and theoretical reconstruction from Paris' law of the parameter of crack length a. (c) Localization of specific points on the crack micrograph.

evaluate the magnitude of the stress intensity factor ΔK using Griffith's fracture theory. The relationship between ΔK and the crack length a was assumed to have the form $\frac{\Delta K}{2} = \frac{EU}{\sqrt{a}} g\left(\frac{a}{w}\right)$, where U is the amplitude of the vibration applied to the end of the specimen, w is its width, and g is an unknown function depending on the ratio $\frac{a}{w}$.[1] Harmonic calculations allowed us to evaluate g for different crack lengths. The results in Fig. 3(a) highlight a linear relationship, expressed as $g\left(\frac{a}{w}\right) = \alpha \times \frac{a}{w}$, from which a simple linear regression gives $\alpha = 0.7002$. Paris' law is considered for crack propagation,

$$\frac{da}{dN} = C\Delta K^n, \tag{2}$$

where N is the number of cycles, and C and n are material coefficients. Using the above relations, Eq. (2) can be integrated over time to give an expression for the crack length,

$$a = \left(a_0^{1-\frac{n}{2}} + \left(1-\frac{n}{2}\right) fC\left(\frac{\alpha EU}{w}\right)^n (t-t_0)\right)^{\frac{1}{1-\frac{n}{2}}}, \tag{3}$$

where t_0 is the time at which the crack length is equal to a_0. The parameters C and n are determined by minimizing the discrepancy between the experimental data and the model obtained using Eq. (3). A limited time range, from 100.0 to 351.5 s, is considered, and the optimum is obtained for $C = 1.05 \times 10^{-7}$ mm/cycle/$(\text{MPa}\sqrt{\text{m}})^n$ and $n = 3.46$.

For these values, the overall shape of the curve given by the experimental data is well represented [Fig. 3(b)]. The ranges for the crack growth rate and the stress intensity factor are $[7.3 \times 10^{-8}; 1.2 \times 10^{-5}]$ mm/cycle and $[1.9; 8.3]$ MPa $\cdot \sqrt{\text{m}}$, respectively. Some deviations can be identified, where the crack propagates either slower or quicker than predicted. From the crack micrography in Fig. 3(c), these particular points cannot be associated with a bifurcation or an abrupt change in the crack's propagation direction. Further analysis is necessary to establish a link with the microstructure, such as a grain boundary.

This study developed an experimental setup that allows the real-time monitoring of crack propagation during ultrasonic fatigue testing. Based on ultrasonic imaging of the surface of the investigated specimen, this method takes advantage of the strong sensitivity of ultrasound to cracks. The use of a PA probe combined with a PWI strategy allowed us to obtain both high spatial and temporal resolutions, allowing us to accurately resolve the crack dynamics. Under these conditions, the full experiment can be performed over the course of a couple of minutes, whereas several hours may be required for more common approaches with sparse sampling of the propagation. Further development could include switching to a laser ultrasound approach for the Rayleigh waves,[24–26] notably to investigate environmental effects. This would allow higher-frequency waves to be generated, therefore improving the spatial resolution while maintaining the same temporal resolution. It would then be possible to investigate crack propagation on both a macroscopic and a mesoscopic scale, leading to a better physical understanding of the associated physical phenomena in complex environments. More complex data analysis methods could also be employed to extract the crack front, such as machine learning or digital image correlation. Dedicated libraries could also be considered, such as those proposed in the literature.[27]

See the supplementary material for a video showing the time evolution of Fig. 2.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

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Author Contributions

R. Delalande: Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Supervision (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **M. Ducousso**: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **E. Monteiro**: Formal analysis (equal); Software (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **N. Ranc**: Conceptualization (equal); Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon request.

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